

FairLight: City-Wide Emergency Vehicle Route Prioritization via Envy-Free Hierarchical Multi-Agent Actor–Critic

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Abstract

Urban traffic signal control is a complex sequential decision-making problem in which decisions made at one intersection can influence traffic conditions at neighboring and downstream intersections. The problem becomes increasingly challenging as the size of the road network grows because a controller must simultaneously achieve network-level coordination, local responsiveness, and computational scalability. Reinforcement learning has emerged as a promising approach for adaptive traffic signal control, while multi-agent reinforcement learning (MARL) allows control to be distributed among individual intersections. However, fully decentralized approaches may lack sufficient global context, whereas centralized approaches become difficult to scale.

This paper proposes **FairLight**, a **Hierarchical Region-Aware Multi-Agent Reinforcement Learning framework for Fair and Adaptive Traffic Signal Control**. The framework introduces a regional abstraction between network-level coordination and intersection-level decision-making. The traffic network is partitioned using graph community detection. Regional traffic states are subsequently processed by a high-level Meta-Policy using spatial and temporal representations, while low-level intersection agents use local observations, neighboring information, and regional guidance to select signal actions.

A central feature of the proposed framework is the explicit treatment of **fairness as a traffic-control objective**. Conventional evaluation based only on average delay, average travel time, or throughput can hide excessive delays experienced by a smaller group of vehicles. The proposed framework therefore incorporates vehicle-level waiting-time distribution, maximum waiting time, tail waiting time, and Jain’s fairness index into optimization and evaluation. A multi-objective reward function jointly considers traffic efficiency, fairness, and emergency responsiveness.

The framework also treats **region construction as part of the research problem rather than merely a preprocessing step**. DBSCAN, Louvain, Leiden, and hybrid region construction methods are investigated to determine how the choice of regional abstraction affects downstream traffic-control performance.

Keywords: Traffic Signal Control, Multi-Agent Reinforcement Learning, Emergency Vehicle Routing, Community Detection, Louvain Algorithm, Fairness-Aware Optimization, Hierarchical Reinforcement Learning

1 Introduction

Urban traffic signal control is a complex sequential decision-making problem where decisions at one intersection propagate to neighboring and downstream intersections. Traffic demand varies dynamically with time, incidents, weather, and travel patterns, making static or rule-based controllers insufficient for real-world conditions. Adaptive systems such as SCOOT introduced traffic-responsive coordination [2], while reinforcement learning approaches have since enabled data-driven signal control [11]. Multi-agent reinforcement learning (MARL) further distributes control across intersections, improving scalability, though fully decentralized approaches risk optimizing locally at the expense of network-wide coordination [20, 19].

Beyond aggregate congestion, traffic signal decisions simultaneously affect travel time, delay, queue formation, throughput, fuel consumption, and emissions. Critically, evaluating controllers solely through aggregate metrics such as average waiting time conceals how service is distributed among individual vehicles. Consider two controllers with vehicle waiting times $W_A = [10, 10, 10, 10, 10, 100]$ and $W_B = [25, 25, 25, 25, 25, 25]$: Controller A achieves a lower mean delay yet inflicts severe starvation on one vehicle. This motivates fairness as a first-class control objective rather than a post-hoc evaluation criterion.

A further dimension of real-world uncertainty that existing MARL frameworks largely overlook is the presence of emergency vehicles (EMVs). Most traffic signal control approaches, including hierarchical methods such as HiLight [24] and cooperative approaches such as CoLight [17], assume homogeneous vehicle populations and optimize purely for aggregate efficiency. In practice, ambulances, fire engines, and police vehicles impose stochastic, high-priority demands on the network that cannot be resolved by ordinary signal timing. Delayed EMV passage has measurable consequences: studies report approximately 700 annual fatalities attributable to late ambulance responses, and an estimated annual mean of 4,500 ambulance-involved crashes in the United States [35]. EMVLight demonstrated that jointly optimizing EMV routing and adaptive signal control within a multi-agent actor-critic framework substantially reduces emergency response times [31], but does not address hierarchical city-wide coordination or fairness. These considerations establish EMV responsiveness as a necessary component of any realistic traffic management framework, not an optional extension. At the same time, providing unrestricted priority to an EMV may excessively delay ordinary traffic, potentially transferring congestion to other parts of the network rather than resolving it. EMV prioritization should therefore not be treated as an isolated intersection-level problem: it requires coordination across the surrounding traffic network while maintaining acceptable service for non-emergency vehicles.

Hierarchical reinforcement learning addresses the scalability–coordination tradeoff by separating long-horizon strategic decisions from short-horizon local control [22, 23]. HiLight is particularly relevant here, employing a Meta-Policy with Transformer-based spatial and LSTM-based temporal modelling to coordinate decentralized Sub-Policy agents across large networks [24]. However, its region partitioning is treated as a preprocessing artifact rather than a control-relevant variable, and it neither incorporates EMV priority nor vehicle-level fairness.

This paper proposes **FairLight: City-Wide Emergency Vehicle Route Prioritization**, an **Envy-Free Hierarchical Multi-Agent Actor-Critic** framework that jointly addresses these limitations. The framework partitions the traffic network using graph community detection methods (DBSCAN, Louvain, Leiden, and hybrid variants), treating region construction as a research variable whose effect on downstream control performance is explicitly investigated. A hierarchical coordination layer issues regional guidance to local intersection agents, while a multi-objective reward function simultaneously optimizes traffic efficiency, vehicle-level fairness, and EMV responsiveness. To the best of our knowledge, no existing framework jointly integrates hierarchical MARL, topology-aware regional partitioning, envy-free fairness optimization, and dynamic multi-EMV routing within a unified city-wide architecture.

2 Related Work

Conventional traffic signal control has primarily relied on analytical optimization and rule-based methods for determining optimal signal timings. Webster’s signal timing formulation introduced one of the earliest mathematical models for fixed-cycle traffic signal optimization, while later adaptive traffic control systems such as SCOOT and SCATS dynamically adjusted signal timings using detector-based traffic measurements and predefined control strategies [1, 2, 3, 4]. Max-pressure control further improved network throughput by selecting signal phases according to queue pressure differences between incoming and outgoing links [5]. Although these approaches are computationally efficient and have been successfully deployed in practice, they rely heavily on predefined traffic models and handcrafted optimization rules. Consequently, they struggle to adapt to highly dynamic urban environments involving unpredictable congestion, heterogeneous intersections, accidents, and emergency vehicle requests.

To improve adaptability, several optimization and machine learning techniques have been introduced for traffic signal control. Evolutionary optimization methods such as Genetic Algorithms optimize signal timing plans through iterative search, while Fuzzy Logic Controllers employ expert-defined rules to respond to uncertain traffic conditions [6, 7]. Later studies adopted machine learning techniques including Support Vector Machines, Artificial Neural Networks, and Long Short-Term Memory (LSTM) networks to predict traffic flow and congestion using historical observations [8, 9, 10]. Although these methods improve prediction accuracy and optimization quality, they primarily estimate future traffic states rather than learning sequential traffic control policies. Furthermore, they require extensive historical training data and cannot continuously adapt to rapidly changing traffic conditions.

Recent advances in deep reinforcement learning have significantly improved adaptive traffic signal control

by enabling traffic controllers to learn directly through interaction with the traffic environment. IntelliLight demonstrated the effectiveness of Deep Q-Networks for adaptive traffic signal control at isolated intersections [11]. PressLight integrated max-pressure traffic theory into deep reinforcement learning to maximize traffic throughput while reducing handcrafted reward engineering [12]. FRAP modeled traffic control as a phase competition problem and exploited traffic phase symmetry to improve learning efficiency [13]. AttendLight later introduced an attention-based architecture capable of generalizing across heterogeneous intersections with varying lane configurations [14]. MetaLight focused on transferring learned policies across unseen intersections [15], while Advanced-XLight demonstrated that effective traffic state representation can significantly influence performance in reinforcement-learning-based traffic signal control [16]. Despite these advances, existing deep reinforcement learning methods primarily optimize local traffic efficiency and do not jointly consider emergency vehicle prioritization, fairness, or large-scale hierarchical coordination.

To improve cooperation among multiple intersections, graph neural networks and multi-agent reinforcement learning have received increasing attention. CoLight introduced graph attention networks to enable communication among neighboring intersections for cooperative traffic signal control [17]. MPLight further improved scalability by introducing parameter sharing across reinforcement learning agents, allowing efficient deployment over city-scale transportation networks [18]. HiLight extended this concept by proposing a hierarchical reinforcement learning framework consisting of Meta-Policy and Sub-Policy agents for coordinating thousands of intersections efficiently [24]. CityFlow provides a large-scale multi-agent reinforcement learning environment designed specifically for city traffic scenarios and supports evaluation of scalable traffic-control algorithms [21]. Although these methods substantially improve scalability and inter-agent cooperation, their primary objective remains maximizing traffic efficiency. They neither address emergency vehicle routing nor incorporate fairness-aware resource allocation across different regions of a city.

HiLight’s hierarchical Meta-Policy/Sub-Policy formulation [24], like most of the methods discussed above, operates under a closed-world assumption: it is trained and evaluated on recurrent congestion patterns and does not model non-recurrent disruptions such as emergency vehicle preemption or traffic incidents. This omission is not merely conceptual. Poorly coordinated EMV preemption has been shown to increase average delay for cross-traffic by 18–32 seconds per vehicle per preemption event, with these localized delays cascading into network-wide congestion [32]. Even well-established preemption strategies carry a measurable efficiency cost: EMVLight’s own evaluation shows that traditional preemption techniques increase average non-EMV travel time by approximately 10% relative to unpreempted baselines [31]. These results indicate that emergency vehicle disruption is a first-order cost of operating a real traffic network, not a negligible edge case that a closed-world controller can safely ignore. Motivated by this, our framework treats EMVs and accident events as first-class constraints on the control problem rather than external events layered on top of it, at the cost of an increase in average travel time and delay relative to a disruption-free benchmark—a cost we treat as the price of a tighter approximation to real-world operating conditions.

Emergency vehicle routing and traffic signal pre-emption have also received considerable research attention. Conventional emergency routing methods generally employ shortest-path algorithms such as Dijkstra’s algorithm, A*, and various time-dependent shortest path formulations to determine optimal routes under static or partially dynamic traffic conditions [25, 26, 27, 28, 29]. More recent studies have introduced reinforcement learning-based routing strategies capable of adapting to changing traffic environments [30]. EMVLight jointly optimizes emergency vehicle routing and adaptive traffic signal control using multi-agent actor–critic reinforcement learning, significantly reducing emergency response times while maintaining acceptable traffic efficiency for ordinary vehicles [31]. Subsequently, hierarchical multi-agent actor–critic methods have been proposed to further improve emergency routing through coordinated traffic signal control [33]. However, existing emergency traffic management approaches generally assume decentralized coordination and do not explicitly consider fairness, topology-aware regional decomposition, or simultaneous optimization for multiple emergency vehicles operating within large urban transportation networks. Visual sensing-based approaches have further demonstrated that incorporating real-time distance measurement and vehicle counting into traffic management can substantially improve EMV scheduling and intersection clearance [35].

Fairness has recently emerged as an important objective in intelligent transportation systems. FairMARL introduced a fairness-aware multi-agent reinforcement learning framework based on generalized social welfare functions to reduce disparities among intersections while maintaining competitive traffic efficiency [34]. Although fairness-aware optimization improves equitable resource allocation, existing approaches primarily

focus on balancing intersection delays and do not integrate emergency vehicle routing, hierarchical coordination, or topology-aware traffic management into a unified framework.

Overall, significant progress has been achieved in analytical optimization, deep reinforcement learning, graph-based coordination, emergency vehicle routing, and fairness-aware traffic signal control. However, these objectives have largely been investigated independently. Classical optimization methods lack adaptability, deep reinforcement learning approaches prioritize traffic efficiency over fairness, graph-based coordination methods do not explicitly support emergency response, emergency vehicle routing methods overlook hierarchical city-wide coordination, and fairness-aware methods ignore emergency traffic management.

To the best of our knowledge, no existing framework jointly integrates **hierarchical multi-agent reinforcement learning, topology-aware regional partitioning, envy-free fairness optimization, and dynamic multi-emergency vehicle routing** within a unified city-wide traffic management architecture. The proposed framework, **FairLight**, addresses these limitations by introducing an **Envy-Free Hierarchical Multi-Agent Actor-Critic** architecture that simultaneously optimizes traffic efficiency, fairness, emergency response, and scalable coordination across heterogeneous urban transportation networks.

3 Multi-Objective Reward Function Design

An effective traffic signal controller must optimize multiple objectives simultaneously rather than maximizing traffic throughput alone. Conventional reinforcement learning approaches primarily optimize queue lengths or average waiting times, often producing policies that systematically disadvantage certain intersections, neglect pedestrian requirements, and fail to prioritize emergency vehicles adequately. To address these limitations, we design a multi-objective reward function that jointly optimizes traffic efficiency, fairness, and emergency vehicle priority.

The overall reward at each decision step is defined as

$$R_{\text{total}} = \frac{4}{9}R_{\text{eff}} + \frac{2}{9}R_{\text{fair}} + \frac{3}{9}R_{\text{emergency}}, \quad (1)$$

where each component corresponds to a distinct traffic management objective.

Traffic Efficiency Reward

The efficiency component encourages continuous improvement in network performance by rewarding reductions in congestion rather than rewarding only low congestion states. It is defined as

$$R_{\text{eff}} = -\beta_1\Delta Q - \beta_2\Delta W + L_{\text{avg}}, \quad (2)$$

where ΔQ denotes the change in total queue length between successive decision steps, ΔW represents the change in cumulative vehicle waiting time, and L_{avg} is the average local throughput reward aggregated across all intersections. The coefficients β_1 and β_2 determine the relative importance assigned to queue reduction and waiting-time reduction.

Unlike conventional reward formulations based solely on absolute queue lengths, the use of temporal differences provides denser learning signals by rewarding measurable improvements between consecutive time steps. This enables faster convergence while encouraging policies that continuously reduce congestion.

Fairness Reward

Optimizing only traffic efficiency frequently results in certain approaches receiving excessively long red phases, leading to persistent vehicle starvation. To ensure equitable service across the network, a fairness reward is incorporated:

$$R_{\text{fair}} = -0.3\text{Var}(W) - 0.25\max(W - 90, 0) - 0.15(1 - J) \\ - 0.5\text{Envy} - 0.5\max(W_{\text{ped}}) - 1.2C_{\text{ped}}. \quad (3)$$

where $\text{Var}(W)$ is the variance of vehicle waiting times, J denotes Jain’s Fairness Index, Envy represents the difference between the best-performing neighboring intersection and the current intersection, W_{ped} is pedestrian waiting time, and C_{ped} measures pedestrian-vehicle conflicts.

Each penalty addresses a different source of inequity. Waiting-time variance promotes balanced service among approaches, the starvation penalty prevents excessively long delays, Jain’s Fairness Index encourages equitable signal allocation, the envy term discourages local policies that sacrifice one intersection for another, while pedestrian waiting and conflict penalties ensure that pedestrian mobility and safety are jointly optimized with vehicular traffic.

Emergency Vehicle Reward

Emergency vehicles require substantially higher priority than regular traffic because delays directly affect emergency response times. Their reward component is therefore defined as

$$R_{\text{emergency}} = -W_{\text{emergency}}, \quad (4)$$

where $W_{\text{emergency}}$ is the cumulative waiting time experienced by emergency vehicles. This formulation directly penalizes any delay encountered by ambulances, fire trucks, or police vehicles, encouraging the controller to allocate green phases immediately upon emergency vehicle detection.

Reward Weight Selection

The weighting coefficients $(\frac{4}{9}, \frac{2}{9}, \frac{3}{9})$ reflect the relative importance of each optimization objective. Traffic efficiency receives the highest weight because congestion reduction remains the primary objective of urban traffic signal control. Emergency vehicle priority is assigned the second-highest weight owing to its critical societal importance. Fairness receives the remaining weight to prevent pathological policies that consistently disadvantage particular intersections.

Finally, the combined reward is normalized using an exponential moving average before being supplied to the reinforcement learning algorithm. Reward normalization reduces reward variance, improves numerical stability during training, and accelerates policy convergence.

4 Future Works

4.1 Enhanced Emergency Vehicle Routing

The routing mechanism presented in this work improves on EMVLight’s reactive dynamic Dijkstra heuristic by integrating route planning into the hierarchical Meta-Policy. However, several directions remain open. At each Meta-Policy decision step, a candidate route set is generated from current regional pressure readings and the known destination, and sub-goals are issued to Sub-Policy agents along candidate corridors to reduce queue lengths before EMV arrival. This transforms routing from a two-horizon problem (current link and next link) into a multi-horizon planning problem aligned with the hierarchical structure. Future work will extend this to settings where the destination is unknown at dispatch time—for example, in mass-casualty events where the nearest available hospital is itself dynamically determined—and to scenarios with more than two simultaneously active EMVs whose corridors intersect repeatedly. The Louvain community structure provides a natural abstraction for corridor planning at the boundary level: because inter-community transitions represent sparse bottleneck links, the Meta-Policy can reason about sequences of community transitions rather than individual links, substantially reducing the planning horizon while preserving global routing awareness.

4.2 Envy-Free Traffic Control

The envy-free regularization introduced in this work operates within individual Louvain communities during training. Future work will investigate adaptive scheduling of λ_e across training: starting with a high efficiency weight and gradually increasing the fairness weight as the policy matures may accelerate convergence while preserving equity in the final policy. Additionally, the definition of envy used here is pairwise and local; extending it to group envy—where a set of intersections collectively prefers another group’s allocation—would capture more subtle structural inequities in large networks. The interaction between the envy-free

penalty and the EMV pre-emption reward also warrants further study: during active EMV passage, pre-emption agents and normal agents necessarily receive asymmetric rewards, and calibrating λ_e dynamically based on EMV proximity could allow the system to temporarily suspend fairness constraints along the active corridor while preserving them network-wide.

4.3 Accident-Based Traffic Control on Real City Maps

The accident injection model used in this work assigns a fixed severity parameter κ to each incident and samples accident locations according to a crash probability distribution derived from historical report data. Future work will model accidents with evolving severity: an initial partial blockage ($\kappa < 1$) that transitions to complete blockage ($\kappa = 1$) as secondary vehicles arrive and then gradually clears as the incident is resolved. This temporal evolution of supply disruption is a qualitatively harder problem for the Meta-Policy, requiring it to track incident lifecycle stages rather than treating each accident as a static constraint. Evaluation on the Coimbatore city map will be extended to multi-district scenarios in which accidents in one part of the city cascade congestion into adjacent zones, stressing the inter-community coordination capacity of the Meta-Policy. The combination of accident-induced rerouting, multiple active EMVs, and envy-free fairness enforcement represents the most realistic and demanding test scenario for the complete framework, and constitutes the primary experimental target for the next phase of this research.

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